Examiner only

6. Cars are tested so that the manufacturers know what happens to the passengers during a collision.

In one such test, a car of mass 1 500 kg travelling with an energy of 127 500 J strikes a solid barrier and stops. The force applied to the car by the barrier is 500 000 N.

(a) (i) Name the energy that the car possesses due to its motion. [1]

....

(ii) State the work done by the barrier to stop the car. [1]

Work done = ... J

(b) (i) Use the equation:

$$\text{distance} = \frac{\text{work}}{\text{force}}$$

to calculate the distance over which the force acts. [2]

Distance = ... m

Examiner only

(ii) I. State **one** feature of cars that is designed to keep the driver safer in a head-on collision. [1]

....

II. Explain how this safety feature keeps the driver safer. [2]

....

....

....

7

Turn over.